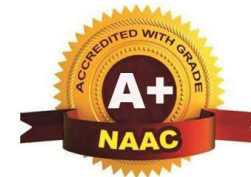




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Dayananda Sagar University (DSU) - Main Campus

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING) Generative -AI (23AM3609)

Synopsis

Title: AI Resume Screening & ATS Scorer

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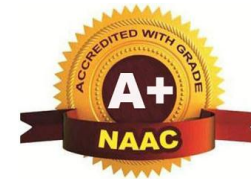
Academic Year: 2025-2026(Even Semester)



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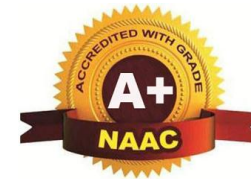
- Aim
- Objectives
- Methodology
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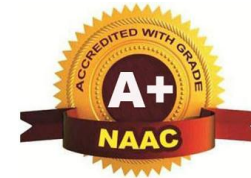


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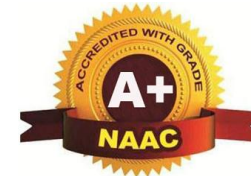
AIM

- To develop an Artificial Intelligence–based recruitment assistance system that automatically analyzes resumes, evaluates candidate suitability for a job role, and generates an ATS compatibility score using Natural Language Processing and Machine Learning techniques.



Objectives

- To design and implement a real-time system capable of recognizing spoken input in multiple languages with high accuracy.
- To extract candidate details such as skills, education, certifications, and experience using NLP techniques.
- To compare resumes with job descriptions and compute an ATS score based on skill matching.
- To rank candidates according to their suitability for a given job role.
- To provide feedback and suggestions for resume improvement using AI.



Methodology

The proposed system follows a modular AI pipeline architecture to automate resume evaluation and candidate ranking. The system integrates resume parsing, Natural Language Processing (NLP), similarity matching, and machine learning-based scoring

A) RESUME UPLOAD AND PREPROCESSING

- 1) The user uploads a resume in PDF or DOCX format
- 2) The system extracts text using document parsers and performs preprocessing .

B) RESUME PARSING AND INFORMATION EXTRACTION

- 1) The extracted text is analyzed using NLP techniques like :
- 2) Named Entity Recognition (NER) , Skill Extraction , Education Detection , Experience Detection , Certificate Detection

C) JOB DESCRIPTION PROCESSING

- 1) The recruiter uploads the job description , the system extracts :
- 2) Required Skills , Experience Level , Preferred qualifications , Keywords

D) SKILL MATCHING AND ATS SCORING

- 1) The system compares the resume with the job description using :
- 2) TF-IDF Vectorization , Cosine Similarity , Keyword weighting , Generates ATS compatibility score (0 – 100 %)

E) CANDIDATE RANKING & FEEDBACK

- 1) Candidates are ranked automatically according to ATS score and the system also identifies :
- 2) Identifies missing skills , Suggests improvements , Provides resume optimization tips

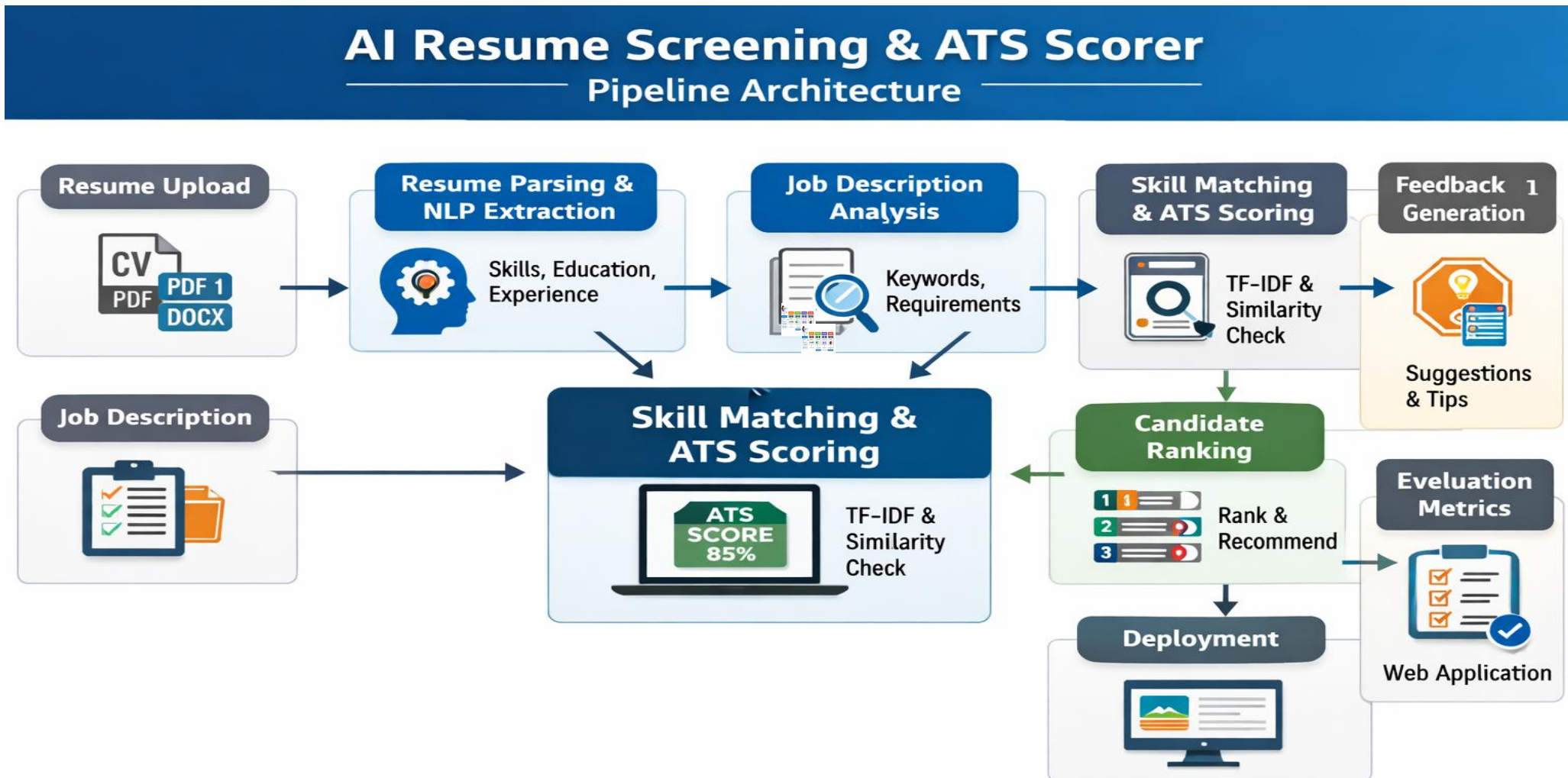
F) EVALUATION AND PERFORMANCE METRICS

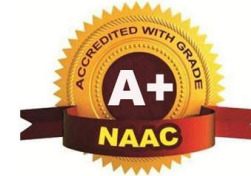
- 1) System performance is evaluated using :
- 2) Accuracy , Precision , Recall , F1-Score , Matching Score Percentage

G) DEPLOYMENT

- 1) The final step will be deployed as a web application using :
- 2) Python , Streamlit , Scikit-learn , NLP libraries

Pipeline Architectural Diagram





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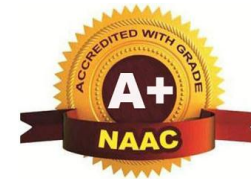
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Thank You!

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